Class 10 Science - Chapter 1

Chemical Reactions and Equations (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)
Class 7-8-9 ki yeh cheezein pehle pakki karo, warna Ch1 samajh nahi aayega.
(Yeh pura RED page hai - revision ke time sabse pehle ise dekho.)

1. Atom, Element, Molecule, Compound

Atom = matter ka sabse chhota kann jo reaction me hissa leta hai.
Element = ek hi type ke atoms se bani cheez. Jaise: Hydrogen (H), Oxygen (O), Iron (Fe), Sodium (Na), Carbon (C).
Molecule = do ya zyada atoms jud kar bante hain. Jaise: O_2, H_2, N_2.
Compound = do ya zyada ALAG elements chemically jud kar bante hain. Jaise: H_2O (paani), CO_2, NaCl (namak).
Yaad rakho: Mixture me cheezein sirf mili hoti hain (alag ki ja sakti), compound me chemically judi hoti hain (aasani se alag nahi hoti).

2. Symbols aur Common Valency (Ratta zaroori)

Symbol = element ka short naam. H, O, C, N, Na, Cl, Ca, Mg, Al, Zn, Cu, Fe, Ag.
Valency = combine karne ki capacity (kitne haath se dosti karta hai).
H = 1, O = 2, Na = 1, Cl = 1, Ca = 2, Mg = 2, Al = 3, Zn = 2.
Common ions: OH^- (hydroxide), CO_3^{2-} (carbonate), SO_4^{2-} (sulphate), NO_3^- (nitrate), HCO_3^- (bicarbonate).
Trick (formula banane ka): valency ko CROSS karke neeche likho.
Na(1) aur SO_4(2) -> Na_2SO_4. Al(3) aur O(2) -> Al_2O_3.

3. Chemical Formula kaise banti hai

Formula batata hai kaunse atoms kitne hain. H_2O = 2 Hydrogen + 1 Oxygen.
Subscript (chhota neeche number) = us atom ki ginti. H_2 me '2' subscript hai.
Agar koi number nahi likha to samjho '1' hai (jaise H_2O me O = 1).

4. Physical Change vs Chemical Change (Bahut important base)

Physical change: sirf roop/shape badalta hai, NAYA padarth nahi banta, aur wapas laaya ja sakta hai. Jaise: ice -> water, paper faadna.
Chemical change: NAYA padarth banta hai, aasani se wapas nahi hota. Jaise: lakdi jalna, doodh ka dahi banna, loha rust hona.
Chemical change ke ishaare (signs): gas nikalna, rang badalna, temperature change, smell change, precipitate (theek neeche baithne wala thos) banna.

5. Reactant aur Product (Reaction ki bhasha)

Reactant = jo cheezein react karti hain (ARROW ke BAAYI taraf - left).
Product = jo cheezein banti hain (ARROW ke DAAYI taraf - right).
Reactants ----> Products
States ke short form: (s)=solid, (l)=liquid, (g)=gas, (aq)=aqueous (paani me ghula hua). Inko likhna marks dilata hai.

6. Law of Conservation of Mass (Class 9 - Ch1 ki jaan)

"Reaction me na to mass banta hai na nasht hota hai."
Matlab: reactant ka total mass = product ka total mass.
Isi wajah se equation BALANCE karni padti hai (dono taraf har atom barabar).

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT
Class 10 Science | Ch 1: Chemical Reactions and Equations

[GREEN = board exam me baar-baar poocha jaata hai]

1. Chemical Reaction kya hai?

Jab ek ya zyada padarth (reactants) milkar naya padarth (product) banate hain,
to use chemical reaction kehte hain.

Chemical reaction ho rahi hai - yeh pehchanne ke 5 signs:
(1) Rang (colour) badalna
(2) State badalna (solid/liquid/gas)
(3) Temperature badalna (garam ya thanda hona)
(4) Gas (bubbles) nikalna
(5) Precipitate (avshep) banna
Memory hook: "Rang-State-Temp-Gas-Precipitate" = "Reaction ka RSTGP signal".

2. Chemical Equation

Reaction ko symbols/formula me likhna = chemical equation.
Example (shabdo me): Magnesium + Oxygen ----> Magnesium oxide
Symbol equation: $2Mg + O_2 \longrightarrow 2MgO$

Balanced Equation: jisme arrow ke dono taraf har element ke atoms barabar hon.
Yeh Law of Conservation of Mass ki wajah se zaroori hai (mass na bane na mite).

3. Equation BALANCE karna (Step-by-step) [VERY IMPORTANT]

Steps (Hit & Trial method):
Step 1: Skeletal (kaccha) equation likho - reactant ----> product.
Step 2: Har element ke atoms dono taraf gino.
Step 3: Sabse zyada atom wale compound se shuru karo, coefficient (bada number aage) laga kar barabar karo. Subscript KABHI mat badlo.
Step 4: H aur O ko aam taur par last me balance karo.
Step 5: Aakhir me states (s, l, g, aq) likho.

Example: $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$
Balanced: $3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$
Yaad rakho: SIRF coefficient badal sakte ho (aage wala number),
subscript (formula ke andar ka number) kabhi nahi.

4. Types of Chemical Reactions (Pura chapter ka dil) [TOP EXAM AREA]

Paanch (5) main types - "CD-DD-Redox" yaad rakho:
Combination, Decomposition, Displacement, Double Displacement, Redox.

(A) Combination Reaction

Do ya zyada reactant milkar EK product banate hain.

General: A + B ----> AB

$CaO + H_2O$ ----> $Ca(OH)_2$ (Quick lime + paani = slaked lime, garmi nikalti)

$2H_2 + O_2$ ----> $2H_2O$

$C + O_2$ ----> CO_2

Yeh ek EXOTHERMIC reaction ka classic example hai (garmi release hoti hai).

(B) Decomposition Reaction (Combination ka ulta)

EK reactant tootkar do ya zyada product deta hai. AB ----> A + B

Teen prakar (energy ke source se):

(1) Thermal (garmi se): $CaCO_3$ --heat--> $CaO + CO_2$

(limestone se quick lime - cement industry me use hota hai)

(2) Electrolytic (bijli se): $2H_2O$ --electricity--> $2H_2 + O_2$

(3) Photolytic (light se): $2AgCl$ --sunlight--> $2Ag + Cl_2$

(AgCl safed se grey ho jaata - photography me use)

Decomposition ko "ulta combination" bolte hain. Yeh aksar ENDOTHERMIC hoti hai (energy SOKHTI hai - heat/light/electricity chahiye).

(C) Displacement Reaction

Zyada reactive element, kam reactive element ko uski jagah se hata deta hai.

General: A + BC ----> AC + B (A zyada reactive hai)

$Fe + CuSO_4$ ----> $FeSO_4 + Cu$ (loha, copper ko hata deta - neela halka hota)

$Zn + CuSO_4$ ----> $ZnSO_4 + Cu$

Reactivity yaad: K > Na > Ca > Mg > Al > Zn > Fe > Pb > (H) > Cu > Ag > Au

Memory: "Kya Naani Ka Magaz Aluminium Zinc Faad Lega? Hydrogen Copper Silver Gold"

(D) Double Displacement Reaction

Do compounds ke ions AAPAS me jagah badal lete hain.

General: AB + CD ----> AD + CB

$Na_2SO_4 + BaCl_2$ ----> $BaSO_4$ (white ppt) + $2NaCl$

Yahan jo solid neeche baithta hai use PRECIPITATE (avshep) kehte hain.

Precipitation reaction = jis double displacement me insoluble solid bane.

(E) Oxidation, Reduction aur Redox [VERY VERY IMPORTANT]

Oxidation = Oxygen ka JUDNA ya Hydrogen ka NIKALNA.

Reduction = Oxygen ka NIKALNA ya Hydrogen ka JUDNA.

Memory hook "OIL RIG": Oxidation Is Loss (of electrons), Reduction Is Gain (of electrons).

Doosra hook: "LEO ke GER" - Lose Electron = Oxidation, Gain Electron = Reduction.

Redox reaction = jisme oxidation aur reduction DONO ek saath hote hain.

$CuO + H_2$ ----> $Cu + H_2O$

(CuO ka O nikla = reduced ; H_2 me O juda = oxidised)

$ZnO + C$ ----> $Zn + CO$ (ZnO reduced, C oxidised)

Oxidising agent = doosre ko oxidise kare (khud reduce ho).

Reducing agent = doosre ko reduce kare (khud oxidise ho).

5. Exothermic aur Endothermic Reactions

Exothermic = HEAT BAHAR nikalti (release). "Exo = Exit = bahar".
Examples: jalna (combustion), respiration (saans), $CaO + H_2O$, doodh/khana sadna, cement set hona.
Respiration ko exothermic reaction kyun kehte? -> Glucose tootkar energy deta:
$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$ + Energy
Endothermic = HEAT ANDAR sokhti (absorb). Garmi do tabhi chalti hai.
Examples: decomposition of $CaCO_3$, photosynthesis.

6. Oxidation ke Daily-Life Effects (Bahut scoring) [GREEN]

(A) CORROSION (ksharan):
Metal ka apni surface par air/moisture se dheere-dheere kharab hona.
Iron par bhura-laal RUST ($Fe_2O_3.xH_2O$) jamta hai.
Rusting ke liye DONO chahiye: Oxygen (air) + Moisture (paani/nami).
Bachne ke tareeke: painting, oiling/greasing, galvanisation (Zn coating), chrome plating, alloy banana (jaise stainless steel).
Silver kala padta (Ag_2S), Copper par hara (basic copper carbonate).

(B) RANCIDITY (vikritgandhita):
Tel/ghee waale khane oxidise hokar kharab smell/taste dete hain.
Rokne ke tareeke:
- Antioxidants daalna (jaise BHA, BHT)
- Air-tight container me rakhna
- Nitrogen gas bhar dena (chips ke packet me - oxygen hata diya jaata)
- Fridge me rakhna (cold), light se door rakhna.

Memory: "Rancidity = tel-ghee ka oxidation se sadna; Nitrogen pack se rokte."

7. Quick Revision Table - Reaction Types

Type	Pehchaan	Example
Combination	A + B -> AB (jud jaaye)	$CaO + H_2O \rightarrow Ca(OH)_2$
Decomposition	AB -> A + B (toot jaaye)	$CaCO_3 \rightarrow CaO + CO_2$
Displacement	A + BC -> AC + B	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Double Displ.	AB + CD -> AD + CB	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4$..
Redox	Oxidation + Reduction	$CuO + H_2 \rightarrow Cu + H_2O$

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Step-by-step. Pehle mushkil, taaki baad me sab aasaan lage.

Example 1 (HARDEST) - Balancing + type + observation (full board style)

Q: Jab ferrous sulphate ($FeSO_4$) crystals ko garam karte hain to:
(a) equation likho aur balance karo (b) reaction type (c) observation.

Solution:
(a) $2FeSO_4 \xrightarrow{heat} Fe_2O_3 + SO_2 + SO_3$
Balance check: Fe: 2=2, S: 2=2 (SO_2+SO_3), O: 8 = 3+2+3 = 8. Balanced.
(b) Type: Decomposition (thermal) - ek reactant tootkar teen product.
(c) Observation: Green crystals ($FeSO_4.7H_2O$) ka rang badal kar bhura ho jaata, aur gandhi (burnt sulphur) smell wali gas nikalti hai.

Example 2 - Identify oxidised & reduced (Redox)

Q: $MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$. Kaun oxidise, kaun reduce hua?

Solution:
- MnO_2 ka oxygen nikal gaya ($MnCl_2$ bana) -> MnO_2 REDUCED hua.
- HCl ka hydrogen nikal kar Cl_2 bana (oxygen juda nahi, par H hata) -> HCl OXIDISED.
- Oxidising agent = MnO_2 ; Reducing agent = HCl.

Example 3 - Double displacement + precipitate

Q: Lead nitrate + Potassium iodide ka reaction likho aur ppt ka rang batao.

Solution:
$Pb(NO_3)_2 + 2KI \longrightarrow PbI_2$ (yellow ppt) $+ 2KNO_3$
Type: Double displacement (precipitation reaction).
Observation: Peela (yellow) PbI_2 precipitate banta hai.

Example 4 - Why is respiration exothermic?

Q: Respiration ko exothermic reaction kyun kehte hain?

Solution:
Glucose, oxygen ke saath tootkar CO_2 + paani banata hai aur ENERGY release karta hai. Yeh energy hamare body ko garam aur active rakhti hai. Energy bahar aati hai isliye exothermic.
$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$ + Energy

Example 5 (EASIEST) - Balance this simple equation

Q: $H_2 + O_2 \longrightarrow H_2O$ ko balance karo.

Solution:
O dono taraf barabar karo: right me O ke liye 2 chahiye -> $2H_2O$.
Ab H: right me 4 hai -> left me $2H_2$.
Balanced: $2H_2 + O_2 \longrightarrow 2H_2O$

PART D - SELF TEST (EASY -> HARD)

Khud likho, phir niche answer hint se check karo. Yeh "test" tumhe yaad rakhne me sabse zyada help karega (active recall).

EASY (1 mark)

Q1. Combination reaction ki definition + 1 example likho.
Q2. (s), (l), (g), (aq) ka matlab batao.
Q3. Rust ka chemical naam/formula kya hai?
Q4. Decomposition reaction kya hai? Ek line.

MEDIUM (2-3 marks)

Q5. In equations ko balance karo:
  (a) $Na + O_2$ ----> Na_2O
  (b) $Al + Cl_2$ ----> $AlCl_3$
  (c) $Pb(NO_3)_2$ ----> $PbO + NO_2 + O_2$
Q6. Exothermic aur Endothermic me 2 differences + 1-1 example.
Q7. Rancidity kya hai? Rokne ke 3 tareeke likho.
Q8. Displacement aur Double displacement me antar (example sahit).

HARD (3-5 marks, board favourite)

Q9. $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$: reaction type + observation + balance check.
Q10. $CuO + H_2 \rightarrow Cu + H_2O$ me oxidation aur reduction dono dikhao
  (kaun oxidised, kaun reduced, oxidising/reducing agent).
Q11. Corrosion kya hai? Iron rusting ke liye kya-kya zaroori? Bachne ke 3 upaay.
Q12. White silver chloride ko dhoop me rakhne par kya hota hai? Equation +
  reaction type batao. (Photography se connection bhi.)

ANSWER HINTS (PART D)

A1. Do/zyada reactant milke ek product. Eg: $CaO + H_2O \rightarrow Ca(OH)_2$.
A2. s=solid, l=liquid, g=gas, aq=aqueous (paani me ghula).
A3. $Fe_2O_3.xH_2O$ (hydrated iron oxide), bhura-laal.
A4. Ek padarth tootkar 2+ padarth dene wali reaction.
A5. (a) $4Na + O_2 \rightarrow 2Na_2O$ (b) $2Al + 3Cl_2 \rightarrow 2AlCl_3$
  (c) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$
A6. Exo = heat nikle (combustion); Endo = heat soke (photosynthesis).
A7. Tel/ghee ka oxidation se kharab hona; antioxidant, airtight, N_2 packing.
A8. Displacement: A+BC->AC+B ($Fe+CuSO_4$). Double: AB+CD->AD+CB ($Na_2SO_4+BaCl_2$).
A9. Thermal decomposition; green->brown, smelly SO_2/SO_3 gas; (atoms balanced).
A10. CuO reduced (O gaya), H_2 oxidised (O juda); ox.agent CuO, red.agent H_2.
A11. Metal ka air+moisture se kharab hona; iron rust ke liye O_2 + paani;

painting/oiling/galvanisation se bachao.

A12. $2AgCl \xrightarrow{sunlight} 2Ag + Cl_2$; photolytic decomposition; safed se grey.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

Generated by Kiro for Krishna1k
